

A Local-Obstruction Counterexample to Conjecture 000000000008

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Claude (Opus 5)[†]

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Abstract

Conjecture 000000000008 asserts that the prime-value count of a square-free integer-coefficient quadratic f admits a lower bound $c_f x / \log x$ with $c_f > 0$. As stated it omits the condition that no prime divides every value of f , and it is false: $f(n) = n^2 + n + 2$ is irreducible over $\mathbb{Q}$, hence square-free, yet every value is even and its only prime value is $f(0) = 2$. The refutation is formalized in Lean 4 against Mathlib.

1 Statement

Conjecture 000000000008 concerns the number of prime values of a square-free integer-coefficient quadratic polynomial f , and asserts that there is a constant $c_f > 0$ with

$$\Pi_f(x) \geq c_f \frac{x}{\log x} \quad (1)$$

for all large x , where $\Pi_f(x) = \#\{n \leq x : f(n) \text{ is prime}\}$.

The conditions imposed are that f be quadratic, have integer coefficients, and be square-free. What is *not* imposed — and what the classical Bunyakovsky and Hardy–Littlewood formulations do impose — is that no prime divide every value of f . Without it the assertion is false.

2 The counterexample

Proposition 1. *Let*

$$f(n) = n^2 + n + 2.$$

Then f is the value function of the integer quadratic

$$F = X^2 + X + 2,$$

which

1. *has integer coefficients 1, 1, 2 and degree 2;*
2. *has discriminant $1^2 - 4 \cdot 1 \cdot 2 = -7 \neq 0$, so no repeated root;*
3. *is irreducible over $\mathbb{Q}$, hence square-free;*
4. *has positive leading coefficient.*

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Proof. Only irreducibility needs an argument. For every $x \in \mathbb{Q}$,

$$4(x^2 + x + 2) = (2x + 1)^2 + 7 \geq 7 > 0,$$

so F has no rational root. A polynomial of degree two over a field with no root in that field is irreducible; and an irreducible element is square-free. $\square$

So f meets every condition the conjecture states, and the classical positivity and non-degeneracy conditions besides.

Theorem 2. $f(0) = 2$ is the only prime value of f . Consequently $\Pi_f(x) = 1$ for every $x \geq 0$, and no constant $c_f > 0$ satisfies (1).

Proof. Write $f(n) = n(n+1) + 2$. One of $n, n+1$ is even, so $n(n+1)$ is even and $f(n)$ is even for every n . For $n \geq 1$ we have $n \leq n^2$, hence

$$f(n) = n^2 + n + 2 \geq 4.$$

So for $n \geq 1$ the value $f(n)$ is an even integer greater than 2, therefore composite. At $n = 0$ we get $f(0) = 2$, which is prime. Hence $\Pi_f(x) = 1$ for every $x \geq 0$.

Suppose (1) held with some $c_f > 0$ for all $x \geq N$. In the equivalent division-free form this says

$$c_f x \leq \Pi_f(x) \log x = \log x \quad (x \geq N).$$

But $\log x = o(x)$, so $\log x \leq \frac{c_f}{2}x$ for all sufficiently large x . Choosing x at least N , at least 1, and large enough for that estimate gives $c_f x \leq \frac{c_f}{2}x$, hence $c_f x \leq 0$, contradicting $c_f > 0$ and $x \geq 1$. $\square$

3 What the counterexample shows

The obstruction is local, at the prime 2:

$$f(n) \equiv 0 \pmod{2} \quad \text{for every integer } n.$$

It is worth being precise about which hypothesis fails, because f satisfies all the others. It is quadratic, has integer coefficients, has nonzero discriminant, is irreducible over $\mathbb{Q}$ — so it is square-free under either reading of that word — and has positive leading coefficient. The sole failure is the fixed prime divisor 2.

A repaired conjecture must therefore add the Bunyakovsky condition: the values $f(n)$, $n \in \mathbb{N}$, have greatest common divisor 1. Under that condition the assertion becomes the quadratic case of the Bunyakovsky conjecture, which is open — it is not known that any irreducible quadratic takes infinitely many prime values.

Remark 3. Squarefreeness is not what is doing the work here, and cannot be made to do it: one can fail the conjecture with a polynomial that is irreducible, which is strictly stronger than square-free. Nor is reducibility needed; $n(n+1)$ would also give a counterexample, but it is reducible and so a referee could object that “square-free” was meant to exclude it. The polynomial above forecloses that objection.

4 Formal verification

The accompanying Lean 4 project formalizes the refutation against Mathlib. The conditions the conjecture places on f are established *about* f , not as side computations on literals: `F_eval` identifies f with the value function of the polynomial `F`, and every subsequent fact about `F` therefore speaks about f .

- `F_eval` — `F.eval n = f(n)` for every n , where $F = X^2 + X + 2$ over $\mathbb{Q}$.
- `F_natDegree`, `F_coeff_zero/one/two`, `F_discriminant` — degree 2, coefficients 2, 1, 1, and discriminant -7 .
- `F_irreducible`, `F_squarefree` — irreducibility via Mathlib's `irreducible_of_degree_le_three_of_no` and `Squarefree` `F` is Mathlib's `Squarefree`, not a re-definition.
- `f_even`, `f_ge_four`, `f_not_prime` — the elementary facts, with primality Mathlib's `Nat.Prime`.
- `f_prime_iff` — `Nat.Prime(f n)  $\iff$  n = 0`, the complete description of the prime values.
- `primeValueCount_eq_one` — $\prod_f(x) = 1$ for every x , the count taken over all $n \leq x$ including $n = 0$.
- `conjecture_00000000008_false` — $\neg \text{LowerBoundHolds}$, where `LowerBoundHolds` states (1) in its division-free form. The proof uses Mathlib's `Real.isLittleO_log_id_atTop`.

Because the conjectured lower bound is itself stated in the formalization, the final theorem is a formalization of its negation rather than of facts that merely entail it.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000008_false` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.